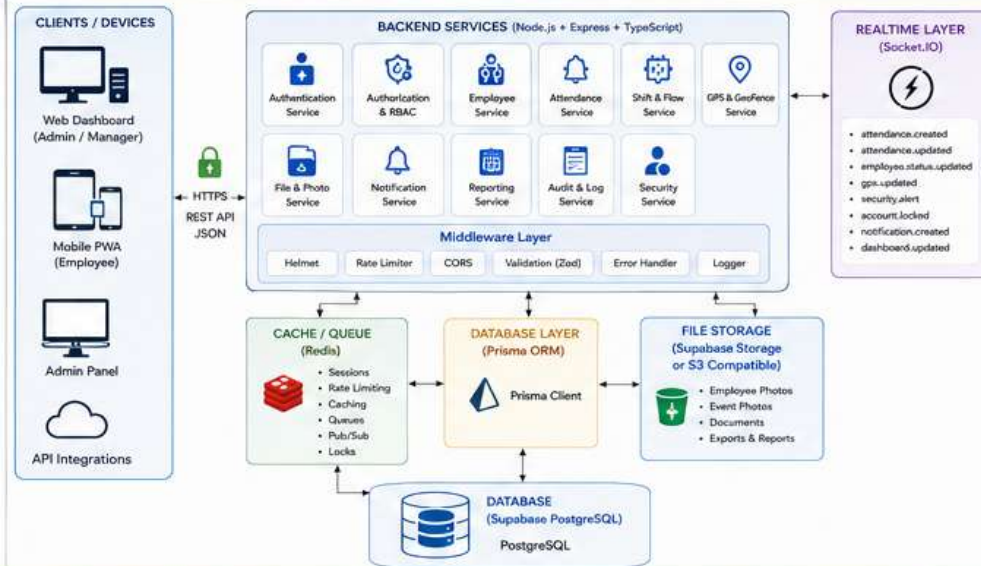
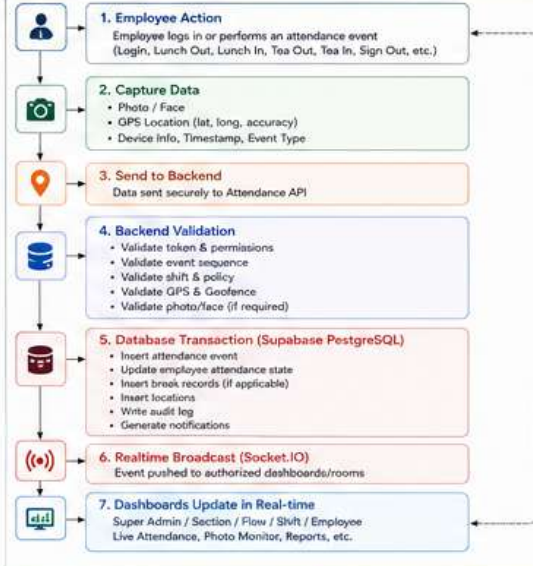


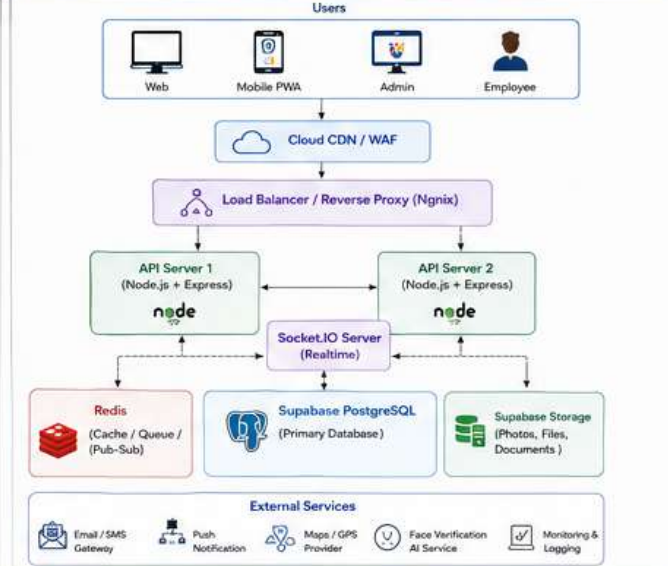
1. HIGH LEVEL SYSTEM ARCHITECTURE



2. ATTENDANCE DATA FLOW (END TO END)



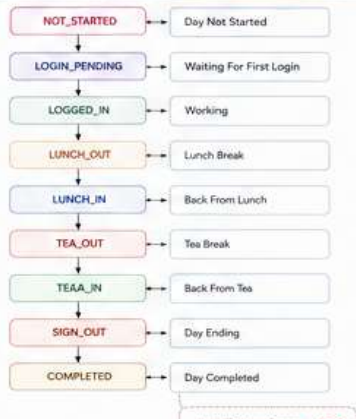
3. SYSTEM DESIGN (DEPLOYMENT VIEW)



4. DATABASE SCHEMA (Supabase PostgreSQL - Key Tables)



5. ATTENDANCE STATE MACHINE (Per Employee Per Day)



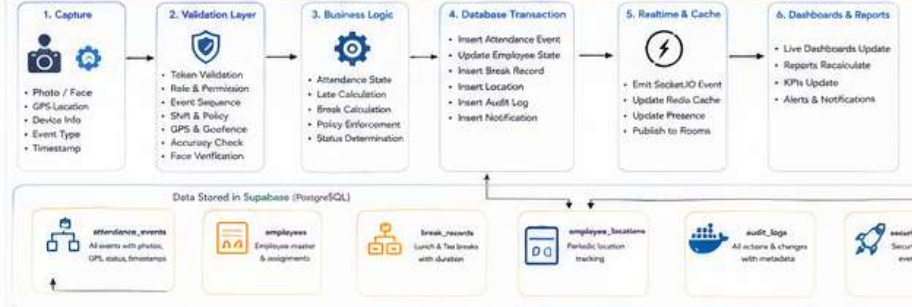
6. MODULES OVERVIEW



7. DASHBOARD SUITE (Real-time)



8. DATA PROCESSING & STORAGE FLOW (Detailed)



9. INFRASTRUCTURE & DEPLOYMENT

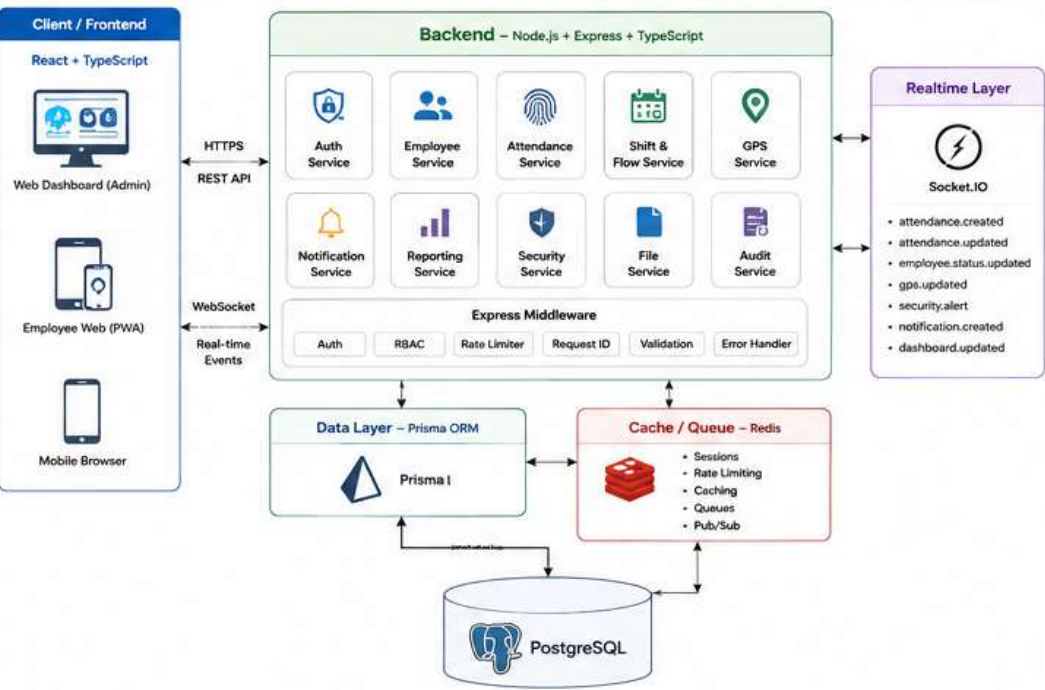


10. PROJECT PLAN / ROADMAP

Phase	Duration	Objectives / Deliverables
Phase 1: Planning & Analysis	1 - 2 Weeks	Requirements, workflows, scope, architecture, tech stack, database design
Phase 2: Core Setup	2 - 3 Weeks	Environment setup, CI/CD, base project, auth, RBAC, Prisma, migrations
Phase 3: Master Data Module	3 - 4 Weeks	Employees, sections, flows, shifts, policies, offices, geofence
Phase 4: Attendance Engine	4 - 5 Weeks	Event engine, state machine, late engine, break engine, validations
Phase 5: GPS & Geofencing	3 - 4 Weeks	GPS capture, accuracy, geofences, location tracking
Phase 6: Dashboards & Realtime	5 - 6 Weeks	All role dashboards, socket.io realtime, live monitors, photo grid
Phase 7: Reports & Analytics	3 - 4 Weeks	Reports, charts, filters, exports (CSV/PDF), summaries
Phase 8: Security & Hardening	2 - 3 Weeks	Lockdown, audit logs, security events, rate limiting, hardening
Phase 9: PWA & Mobile Experience	2 - 3 Weeks	PWA, offline, installable, mobile optimization
Phase 10: Testing & Deployment	2 - 3 Weeks	Unit, integration, E2E tests, performance, deployment

Total Estimated Duration: 26 - 37 Weeks

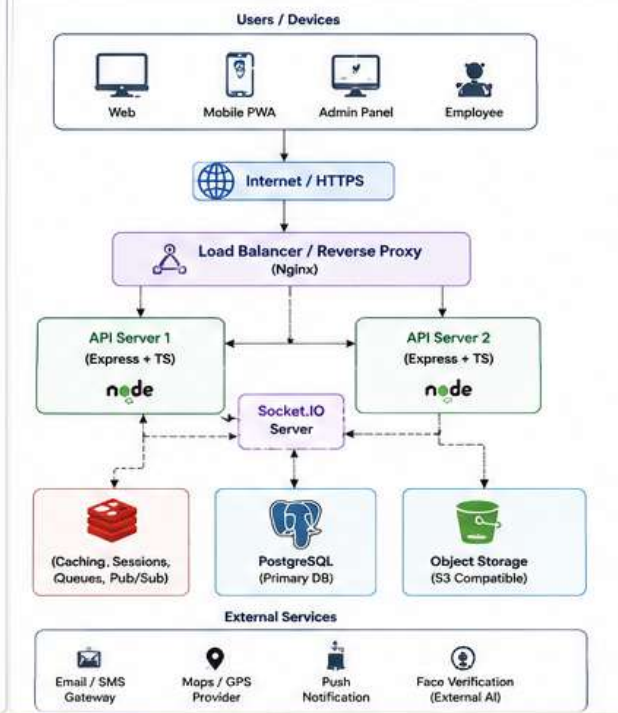
1. SYSTEM ARCHITECTURE DIAGRAM



2. DATA FLOW DIAGRAM (Attendance Event Flow)



3. SYSTEM DESIGN (HIGH LEVEL)

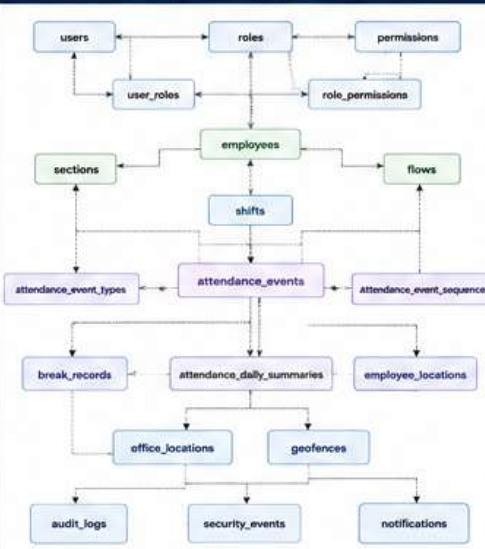


4. PROJECT PLAN & ROADMAP

Phase	Duration	Objectives	Key Deliverables
Phase 1 Planning & Analysis	1-2 Weeks	- Requirement analysis - System design - Technology setup	PRD, FRD, Architecture, ERD, Tech stack setup
Phase 2 Database & Core Setup	2-3 Weeks	- PostgreSQL schema with Prisma - Migrations & seed data - Auth & RBAC system	Database, Auth APIs, Role management
Phase 3 Master Data Modules	3-4 Weeks	- Employee, Section, Flow, Shift - Attendance Policies - Office & Geofence	CRUD APIs + UI for all master modules
Phase 4 Attendance Engine	4-5 Weeks	- Event sequence engine - Attendance creation flow - Late calculation & summaries	Attendance APIs, State machine, Daily summaries
Phase 5 GPS & Geofencing	3-4 Weeks	- Location capture - Geofence validation - Location history	GPS APIs, Map UI, Geofence module
Phase 6 Dashboards & Reports	5-6 Weeks	- All dashboards - Reports & analytics - Export (CSV/PDF)	All dashboards, charts, reports, filters
Phase 7 Notifications & Realtime	2-3 Weeks	- Socket.io integration - Real-time dashboards - Email/SMS notifications	Realtime updates, notifications center
Phase 8 Security & Hardening	2-3 Weeks	- Security events - Account lock, audit logs - Rate limiting, tests	Security modules, audit logs, tests
Phase 9 PWA & Mobile Experience	2-3 Weeks	- PWA setup - Offline handling - Mobile optimization	Installable PWA, mobile ready experience
Phase 10 Testing & Deployment	2-3 Weeks	- Unit, Integration, E2E tests - Performance testing - Docker & deployment	Test reports, Docker images, Production deployment

Total Estimated Duration: 26 - 37 Weeks

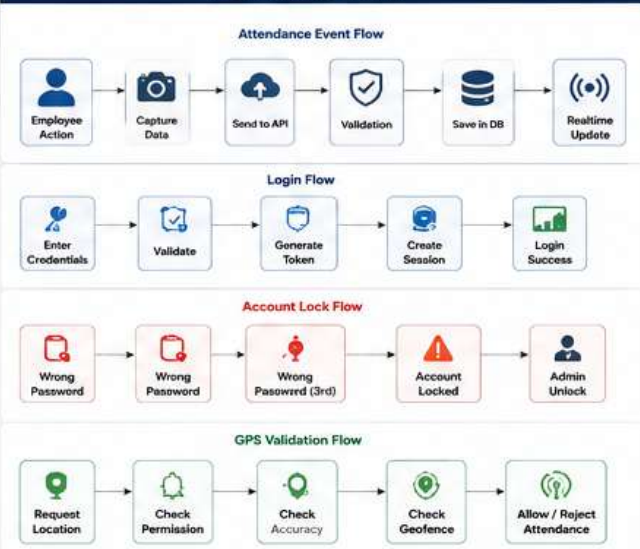
5. DATABASE ERD (Simplified)



6. MODULES OVERVIEW

	Authentication & RBAC Secure login, roles, permissions, sessions
	Employee Management Add, edit, import, manage employees
	Section Management Manage sections & section admins
	Flow Management Manage flows, flow admins, assignments
	Shift Management Six shifts configuration & schedules
	Attendance Engine Event sequence, validation, state machine
	Break Management Lunch & Tea break tracking
	GPS & Geofence Location capture, geofence validation
	Dashboards All role-based dashboards
	Reports & Analytics Charts, reports, export
	Notifications System notifications & alerts
	Security & Audit Security events, audit logs, logout
	System Settings Attendance policies, system configs
	PWA & Mobile Installable PWA, offline support

7. KEY PROCESS FLOWS



TECHNOLOGY STACK SUMMARY

